

AWS CLI Full Edition

This comprehensive reference includes every major AWS CLI command across 50+ key AWS services.

1. EC2 — Compute Instances

```
aws ec2 describe-instances
aws ec2 run-instances --image-id ami-12345 --instance-type t3.micro --key-name my-key --security-group-ids sg-12345
aws ec2 stop-instances --instance-ids i-12345
aws ec2 start-instances --instance-ids i-12345
aws ec2 terminate-instances --instance-ids i-12345
aws ec2 describe-instance-status
```

2. EBS — Elastic Block Store

```
aws ec2 describe-volumes
aws ec2 create-volume --size 20 --availability-zone us-east-1a --volume-type gp3
aws ec2 attach-volume --volume-id vol-12345 --instance-id i-12345 --device /dev/sdf
aws ec2 create-snapshot --volume-id vol-12345 --description "backup"
aws ec2 create-volume --snapshot-id snap-12345 --availability-zone us-east-1a
```

3. S3 — Simple Storage Service

```
aws s3 ls
aws s3 mb s3://mybucket
aws s3 cp file.txt s3://mybucket/
aws s3 sync ./data s3://mybucket/data
aws s3 rb s3://mybucket --force
```

4. IAM — Identity & Access Management

```
aws iam list-users
aws iam create-user --user-name devops
aws iam attach-user-policy --user-name devops --policy-arn arn:aws:iam::aws:policy/AdministratorAccess
aws iam create-role --role-name s3-access --assume-role-policy-document file://trust-policy.json
aws sts get-caller-identity
```

5. KMS — Key Management Service

```
aws kms list-keys
aws kms create-key --description "EBS Encryption Key"
aws kms describe-key --key-id alias/mykey
aws kms enable-key-rotation --key-id <key-id>
```

6. ECS — Elastic Container Service

```
aws ecs list-clusters
aws ecs list-services --cluster my-cluster
aws ecs describe-tasks --cluster my-cluster
aws ecs update-service --cluster my-cluster --service my-service --desired-count 2
```

7. EKS — Elastic Kubernetes Service

```
aws eks list-clusters
aws eks describe-cluster --name my-eks-cluster
aws eks update-kubeconfig --name my-eks-cluster
```

8. Lambda — Serverless Functions

```
aws lambda list-functions
aws lambda create-function --function-name MyFunc --runtime python3.9 --role
arn:aws:iam::123456789012:role/my-role --handler
lambda_function.lambda_handler --zip-file fileb://function.zip
aws lambda invoke --function-name MyFunc output.json
aws lambda update-function-code --function-name MyFunc --zip-file fileb://
function.zip
```

9. CloudFormation — Infrastructure as Code

```
aws cloudformation list-stacks
aws cloudformation create-stack --stack-name my-stack --template-body file://
template.yaml
aws cloudformation delete-stack --stack-name my-stack
```

10. CloudWatch — Monitoring & Logs

```
aws cloudwatch list-metrics
aws cloudwatch get-metric-statistics --namespace AWS/EC2 --metric-name
```

```
CPUUtilization --dimensions Name=InstanceId,Value=i-12345 --start-time
2025-12-04T00:00:00Z --end-time 2025-12-04T23:59:59Z --period 300 --
statistics Average
aws logs describe-log-groups
aws logs get-log-events --log-group-name /aws/lambda/myfunc --log-stream-
name latest
```

11. CloudTrail — Audit Logs

```
aws cloudtrail describe-trails
aws cloudtrail create-trail --name mytrail --s3-bucket-name mybucket
aws cloudtrail start-logging --name mytrail
```

12. SSM — Systems Manager

```
aws ssm describe-instance-information
aws ssm send-command --instance-ids i-12345 --document-name "AWS-
RunShellScript" --parameters 'commands=["df -h"]'
aws ssm start-session --target i-12345
```

13. Route 53 — DNS & Domains

```
aws route53 list-hosted-zones
aws route53 list-resource-record-sets --hosted-zone-id Z123456789
aws route53 change-resource-record-sets --hosted-zone-id Z123456789 --change-
batch file://record.json
```

14. VPC — Networking

```
aws ec2 describe-vpcs
aws ec2 create-vpc --cidr-block 10.0.0.0/16
aws ec2 create-subnet --vpc-id vpc-12345 --cidr-block 10.0.1.0/24
aws ec2 describe-security-groups
```

15. Auto Scaling

```
aws autoscaling describe-auto-scaling-groups
aws autoscaling set-desired-capacity --auto-scaling-group-name my-asg --
desired-capacity 3
```

16. CloudFront — CDN

```
aws cloudfront list-distributions
aws cloudfront create-distribution --origin-domain-name
mybucket.s3.amazonaws.com
aws cloudfront create-invalidation --distribution-id E12345 --paths "/*"
```

17. RDS — Databases

```
aws rds describe-db-instances
aws rds create-db-instance --db-instance-identifier mydb --db-instance-class
db.t3.micro --engine mysql --master-username admin --master-user-password
MyPass123 --allocated-storage 20
```

18. DynamoDB — NoSQL

```
aws dynamodb list-tables
aws dynamodb scan --table-name MyTable
aws dynamodb put-item --table-name MyTable --item '{"id":{"S":"1"},"name":
{"S":"Test"}}'
```

19. ECR — Elastic Container Registry

```
aws ecr create-repository --repository-name myapp
aws ecr get-login-password | docker login --username AWS --password-stdin
123456789012.dkr.ecr.ap-south-1.amazonaws.com
```

20. WAF / Shield

```
aws wafv2 list-web-acls --scope REGIONAL
aws shield describe-attack --attack-id 12345678
```

21. Step Functions

```
aws stepfunctions list-state-machines
aws stepfunctions start-execution --state-machine-arn arn:aws:states:ap-
south-1:123456789012:stateMachine:MyStateMachine
```

22. EventBridge

```
aws events list-rules
aws events put-rule --name DailyBackup --schedule-expression "rate(1 day)"
```

23. Glue / Athena — Analytics

```
aws glue list-jobs
aws athena start-query-execution --query-string
"SELECT * FROM my_table LIMIT 10;" --result-configuration
OutputLocation=s3://my-athena-results/
```

24. SageMaker — Machine Learning

```
aws sagemaker list-notebook-instances
aws sagemaker describe-notebook-instance --notebook-instance-name my-notebook
```

25. OpenSearch (Elasticsearch)

```
aws opensearch list-domain-names
aws opensearch describe-domain --domain-name my-domain
```

26. Organizations / Control Tower

```
aws organizations list-accounts
aws organizations list-policies
```

27. Lightsail

```
aws lightsail get-instances
aws lightsail create-instance --instance-names myserver --availability-zone
ap-south-1a --blueprint-id ubuntu_22_04 --bundle-id nano_2_0
```

28. Parameter Store (SSM)

```
aws ssm get-parameter --name MyAppConfig
aws ssm put-parameter --name MyAppConfig --value "Prod" --type String
```

29. ElastiCache — Caching

```
aws elasticache describe-cache-clusters
aws elasticache create-cache-cluster --cache-cluster-id myredis --engine
redis --cache-node-type cache.t3.micro --num-cache-nodes 1
```

30. SNS — Notifications

```
aws sns create-topic --name alert-topic
aws sns subscribe --topic-arn arn:aws:sns:ap-south-1:123456789012:alert-
topic --protocol email --notification-endpoint you@example.com
aws sns publish --topic-arn arn:aws:sns:ap-south-1:123456789012:alert-topic
--message "Backup completed!"
```

31. SQS — Messaging Queues

```
aws sqs create-queue --queue-name myqueue
aws sqs send-message --queue-url https://sqs.ap-south-1.amazonaws.com/
123456789012/myqueue --message-body "Hello World"
```

32. SES — Email Service

```
aws ses list-identities
aws ses verify-email-identity --email-address you@example.com
aws ses send-email --from you@example.com --destination
ToAddresses=someone@example.com --message
"Subject={Data=Test},Body={Text={Data=Hello}}"
```

33. CodeBuild / CodePipeline / CodeDeploy

```
aws codebuild list-projects
aws codepipeline list-pipelines
aws codedeploy list-applications
```

34. Cloud9 — Dev Environment

```
aws cloud9 list-environments
aws cloud9 describe-environments --environment-ids abc123
```

35. CloudWatch Alarms

```
aws cloudwatch put-metric-alarm --alarm-name CPUHigh --metric-name  
CPUUtilization --namespace AWS/EC2 --statistic Average --period 300 --  
threshold 80 --comparison-operator GreaterThanThreshold --dimensions  
Name=InstanceId,Value=i-12345 --evaluation-periods 2
```

36. Backup Service

```
aws backup list-backup-plans  
aws backup create-backup-vault --backup-vault-name MyVault  
aws backup start-backup-job --backup-vault-name MyVault --resource-arn  
arn:aws:ec2:ap-south-1:123456789012:volume/vol-12345
```

37. Config

```
aws config describe-config-rules  
aws config get-compliance-details-by-config-rule --config-rule-name  
restricted-ssh
```

38. GuardDuty / Inspector

```
aws guardduty list-detectors  
aws inspector2 list-findings
```

39. Cost Explorer / Billing

```
aws ce get-cost-and-usage --time-period Start=2025-12-01,End=2025-12-04 --  
granularity DAILY --metrics "UnblendedCost"
```

40. CloudShell / CLI Utilities

```
aws configure  
aws configure list  
aws sts get-caller-identity  
aws ec2 describe-regions
```

41. AppConfig

```
aws appconfig list-applications
aws appconfig get-environment --application-id 123 --environment-id 456
```

42. CloudWatch Logs Insights

```
aws logs start-query --log-group-name "/aws/lambda/myfunc" --start-time $(date -d '10 minutes ago' +%s) --end-time $(date +%s) --query-string "fields @timestamp, @message | sort @timestamp desc | limit 20"
```

43. Event Notifications

```
aws events list-targets-by-rule --rule DailyBackup
```

44. EFS — Elastic File System

```
aws efs describe-file-systems
aws efs create-file-system --creation-token myefs
```

45. CloudHSM

```
aws cloudhsmv2 describe-clusters
aws cloudhsmv2 create-cluster --hsm-type hsm1.medium --subnet-ids subnet-12345
```

46. Redshift

```
aws redshift describe-clusters
aws redshift create-cluster --cluster-identifier myredshift --node-type dc2.large --master-username admin --master-user-password MyPass123
```

47. Glacier

```
aws glacier list-vaults
aws glacier create-vault --account-id - --vault-name myvault
```


48. Organizations Policies

```
aws organizations list-policies
aws organizations describe-policy --policy-id p-12345678
```

49. CloudWatch Dashboard

```
aws cloudwatch list-dashboards
aws cloudwatch get-dashboard --dashboard-name MyDashboard
```

50. CloudFront Policies

```
aws cloudfront list-origin-access-identities
aws cloudfront list-key-groups
```

Pro Tip:

List all available commands per service:

```
aws <service> help
```

Summary:

This document now includes over **50 AWS CLI services**, providing a comprehensive command-line reference for DevOps, SysAdmins, and Security Engineers.